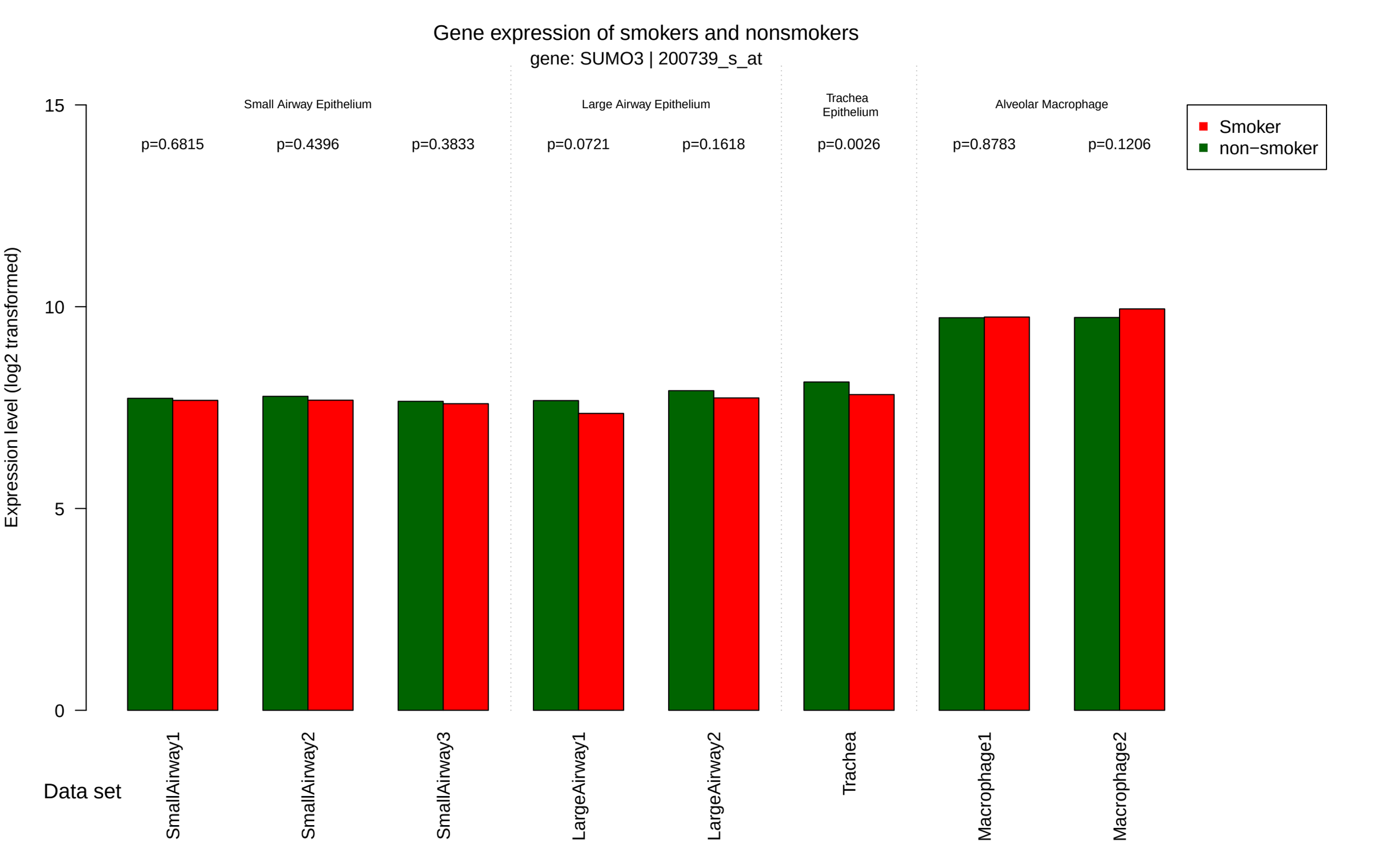
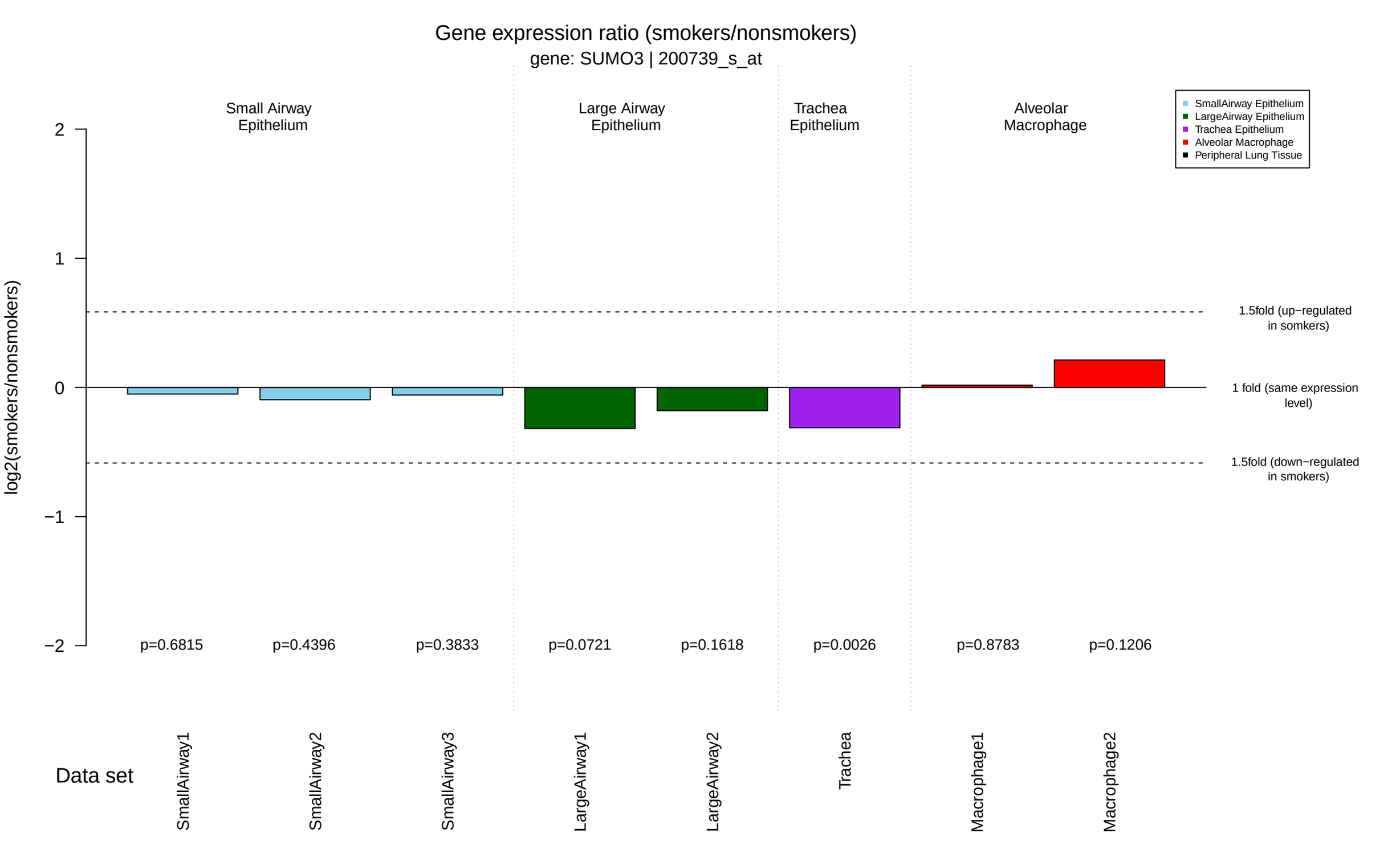
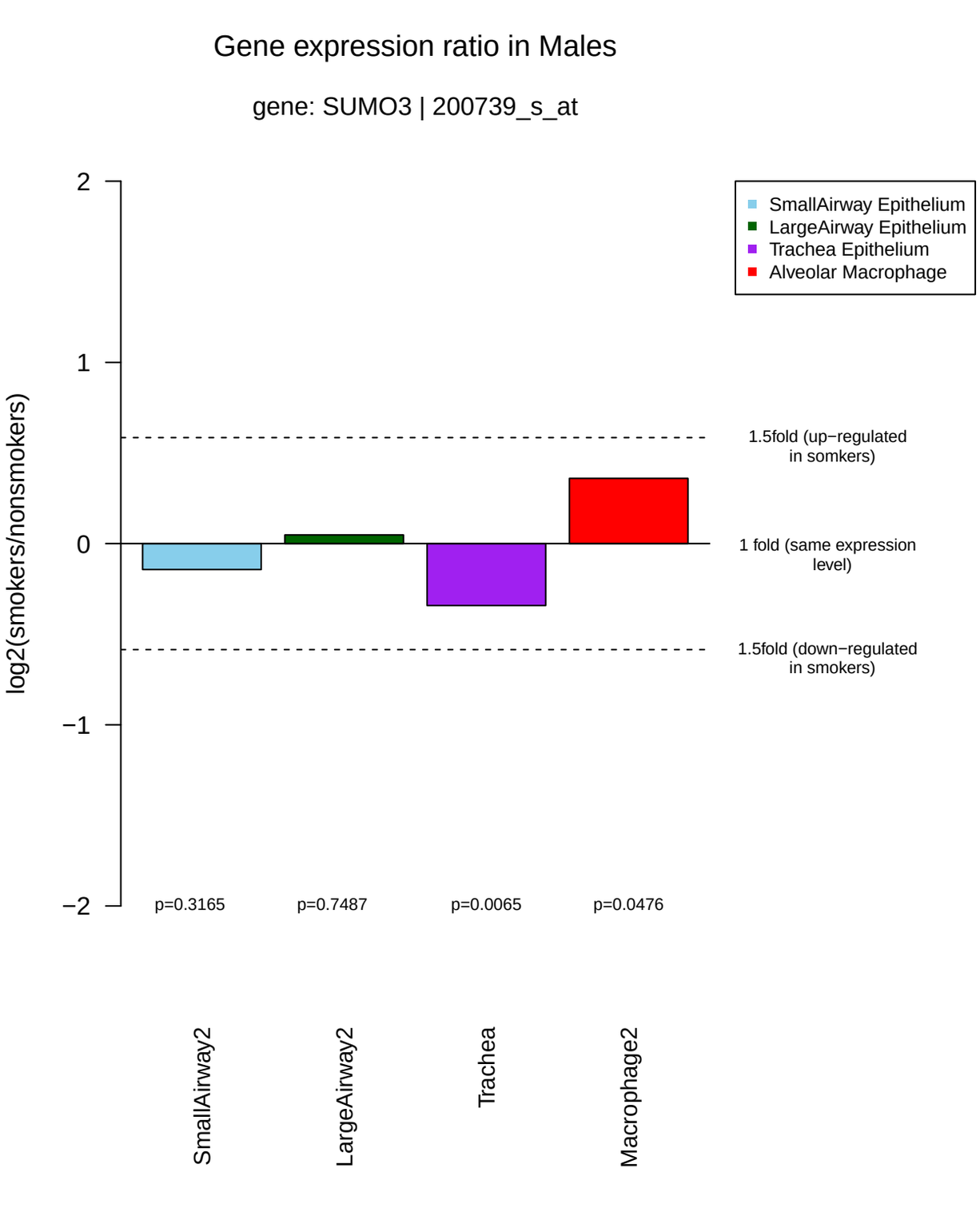
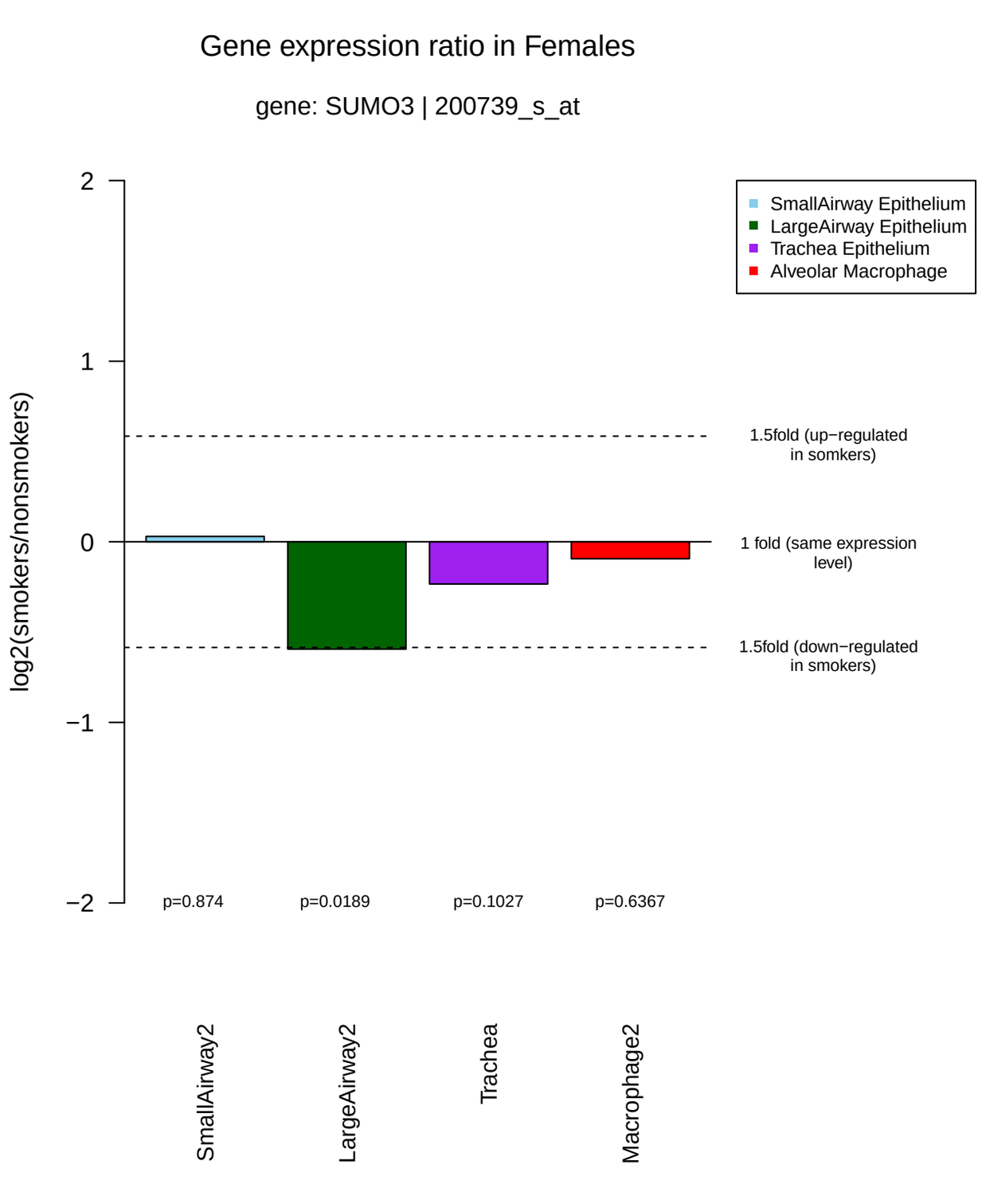
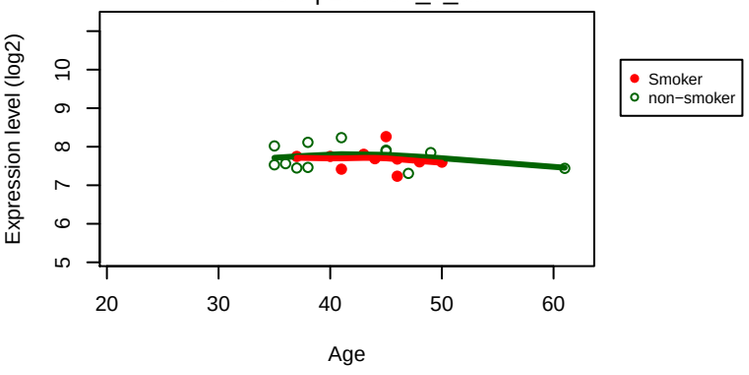
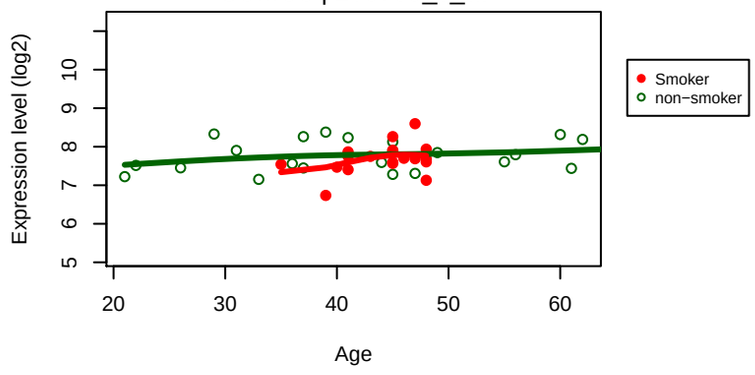

Gene expression of smokers and nonsmokers
gene: SUMO3 | 200739_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6815
p=0.4396
p=0.3833
p=0.0721
p=0.1618
p=0.0026
p=0.8783
p=0.1206
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

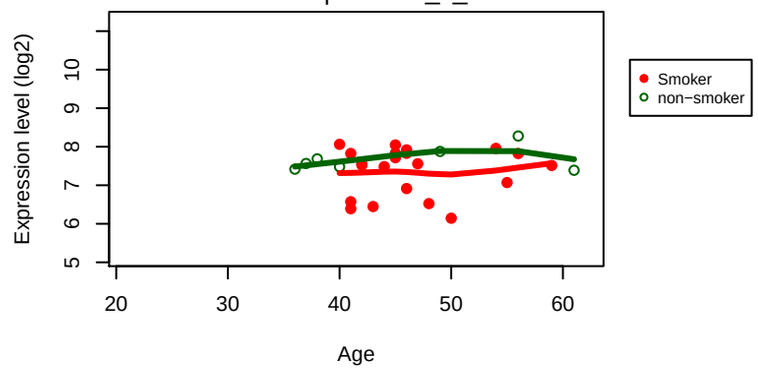

Gene expression ratio (smokers/nonsmokers)
gene: SUMO3 | 200739_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6815
p=0.4396
p=0.3833
p=0.0721
p=0.1618
p=0.0026
p=0.8783
p=0.1206
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

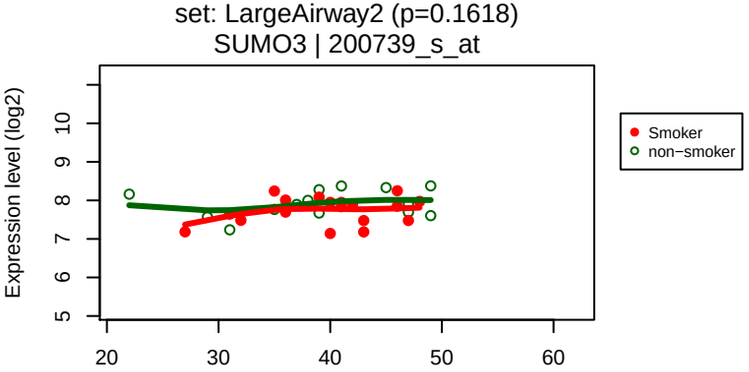

Gene expression ratio in Males
gene: SUMO3 | 200739_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3165
p=0.7487
p=0.0065
p=0.0476
SmallAirway2
LargeAirway2
Trachea
Macrophage2

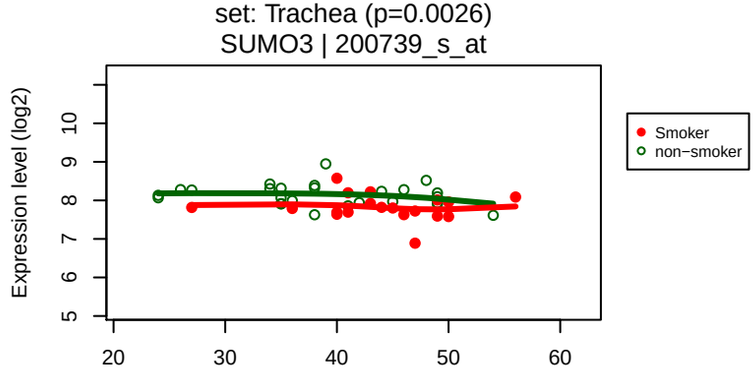

Gene expression ratio in Females
gene: SUMO3 | 200739_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.874
p=0.0189
p=0.1027
p=0.6367
SmallAirway2
LargeAirway2
Trachea
Macrophage2

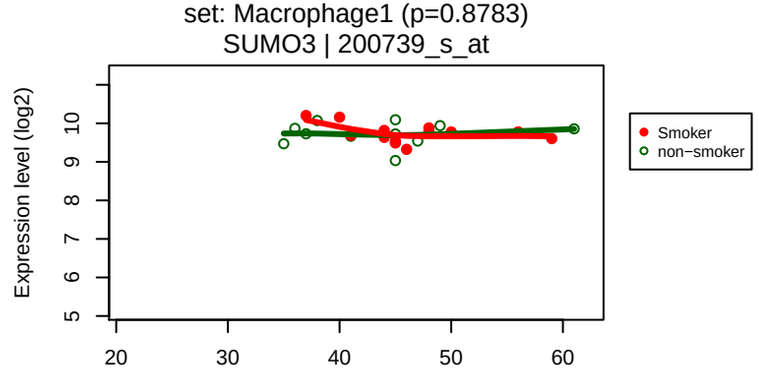

set: SmallAirway1 (p=0.6815)
SUMO3 | 200739_s_at
Expression level (log2)
Age
Smoker
non−smoker

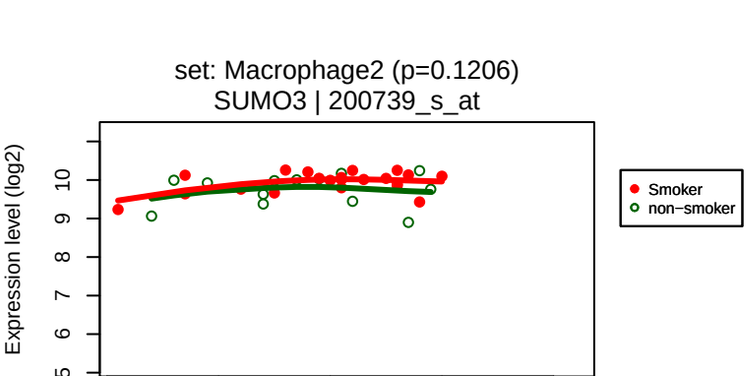

set: SmallAirway2 (p=0.4396)
SUMO3 | 200739_s_at
Expression level (log2)
Age
Smoker
non−smoker

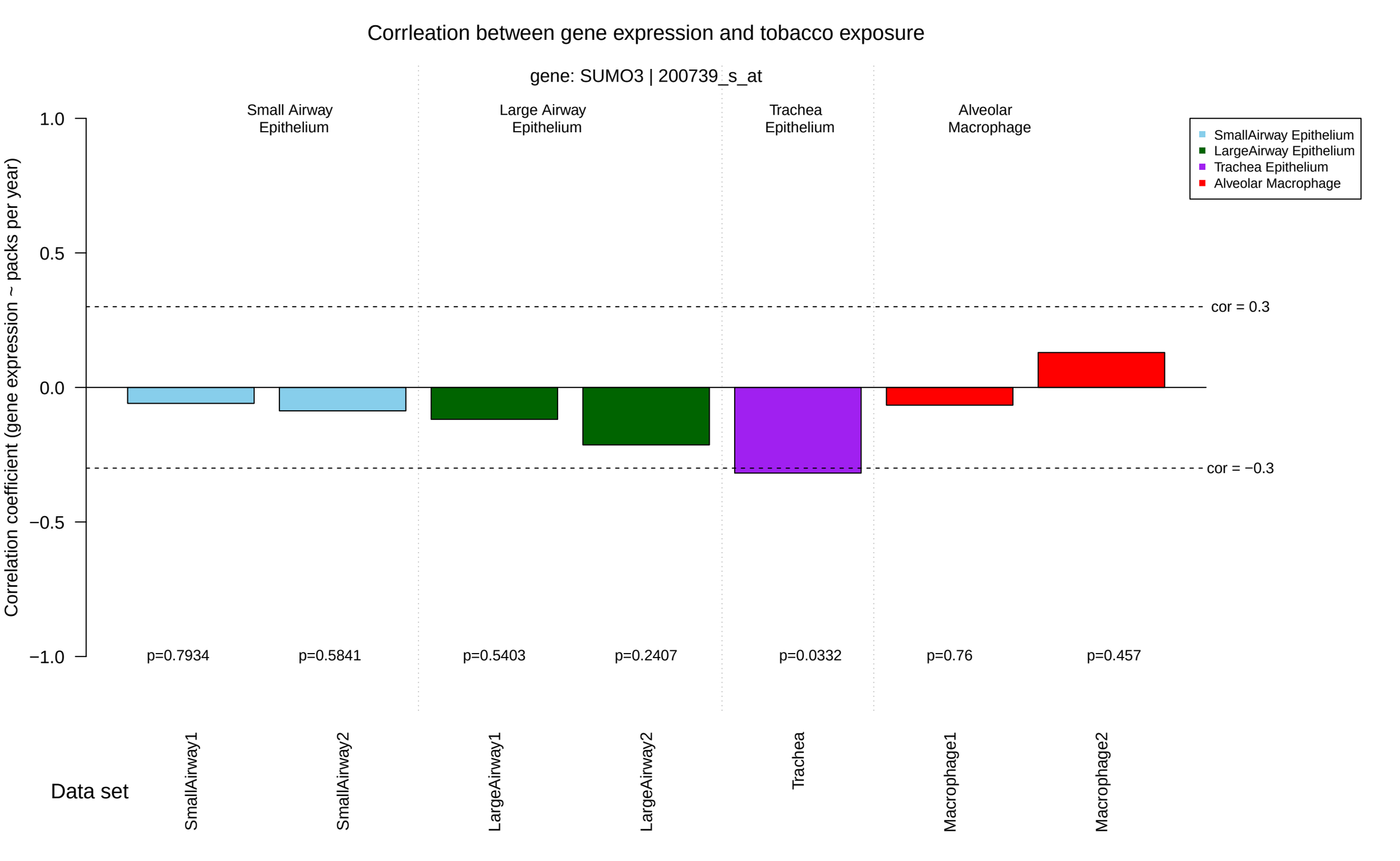

set: LargeAirway1 (p=0.0721)
SUMO3 | 200739_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1618)
SUMO3 | 200739_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0026)
SUMO3 | 200739_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8783)
SUMO3 | 200739_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1206)
SUMO3 | 200739_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SUMO3 | 200739_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7934
p=0.5841
p=0.5403
p=0.2407
p=0.0332
p=0.76
p=0.457
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2